

Credit Card Spending Prediction

The variables in the dataset (summary-data) are related to annual charges by credit card customers.

- AnnualIncome is the customer's annual income
- HouseholdSize is the size of the customer's household
- YrsEdAfterHS is the number of years of education after high school that the customer has
- HrWkTV is the number of hours per week of TV that the customer watches
- AnnualCharges is the total amount of annual credit card charges by the customer

Task 1: Create an appropriate plot to examine the relationship between **Annual Income** and **Annual Charges** (assume that we trying to predict Annual Charges). Comment on the relationship between the variables.

SOLUTION 1:

[PYTHON WORKBOOK](#)



The scatter plot above shows a **positive relationship** between Annual Income and Annual Charges. This means that as income increases, there's a general trend of higher credit card charges. However, the relationship appears to have considerable variability - some customers with similar incomes have vastly different charging patterns.

The **correlation is not perfectly linear**, with points scattered widely around the regression line. This suggests that while income is a predictor of credit card spending, other factors likely influence spending behavior as well.

Task 2: Build a linear regression model with Annual Income to predict Annual Charges.
Comment on the relationship between the variables.

SOLUTION 2:

Simple Linear Regression Model:

As calculated [here](#):

Intercept- 2095.4699:

This is the estimated base annual charge when income is zero.

Coefficient for Annual Income- 135.4137:

For every \$1,000 increase in Annual Income, Annual Charges increase by \$135.41 on average. The positive coefficient confirms that higher income is associated with higher credit card charges

Mean Squared Error: 25736945.8926

R² - 0.4491:

R-squared (R²) value indicates that 44.9% of the variation in Annual Charges is explained by Annual Income. This implies that annual Income alone explains approximately 40-45% of the variation in Annual Charges. So, while income is a significant predictor, about 55-60% of the variation in credit card spending is explained by other factors not captured in this simple model.

NOTE:

A higher R² means a better fit, but it doesn't always mean the model is good (overfitting is a risk).

R² = 1: The model perfectly predicts the data.

R² = 0: The model does not explain any variability in the data.

0 < R² < 1: The model explains some variability, but not all.

Formula: AnnualCharges = 2095.47 + (135.41 × AnnualIncome)

CORRELATION MATRIX

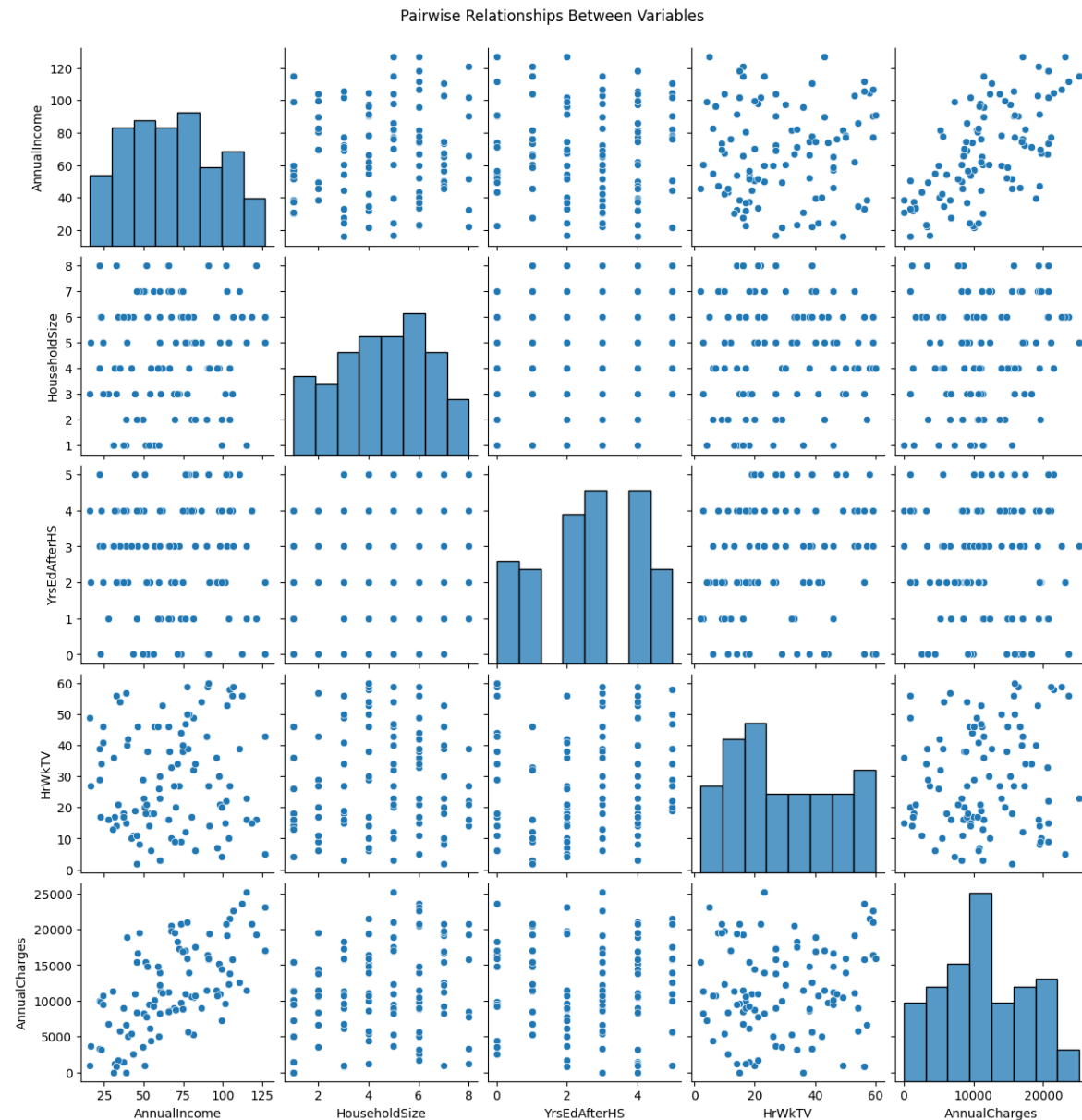
VARIABLE	CORRELATION WITH ANNUAL CHARGES	MEANING
Annual Income	0.634	Strong positive correlation: As income increases, annual charges tend to increase. This makes sense since people with higher incomes likely spend more on credit cards.
HouseholdSize	0.222	Weak positive correlation: Larger households spend slightly more, but the relationship isn't very strong.
HrWkTV	0.116	Very weak positive correlation: Watching more TV has little effect on annual charges.
YrsEdAfterHS	-0.024	Near zero, weak negative correlation: Education level has no meaningful impact on credit card spending.

Task 3: Build the best linear regression model that you can with one or more variables to predict Annual Charges. Comment on the findings.

SOLUTION 3:

To improve my model's performance, I have incorporated all variables (Annual Income, Household Size, Years of Education after High School, and Weekly TV Hours):

- The R-squared (R^2) value increases to **0.5073**, showing that the additional variables help explain more of the variation in Annual Charges, up to 50%.
- Annual Income remains a strong predictor based on coefficient magnitude.
- Household Size shows a positive relationship with Annual Charges, suggesting larger households tend to spend more.
- Years of Education after High School appears to be negatively correlated with spending. This could mean that higher education leads to more financial awareness and thus less credit card spending.
- Hours of TV watched weekly shows a weaker relationship with spending compared to the other variables.



NEW REGRESSION FORMULA:

$$\text{AnnualCharges} = 886.0380 + (128.2698 \times \text{AnnualIncome}) + (520.8074 \times \text{HouseholdSize}) + (-270.7752 \times \text{YrsEdAfterHS}) + (26.5706 \times \text{HrWkTV})$$

Key insights from this analysis:

- Income is the strongest predictor of credit card spending
- Demographic factors like household size and education level provide additional predictive power
- A significant portion of spending variation remains unexplained, suggesting other factors (perhaps psychological, behavioral, or circumstantial) influence credit card charges.